4 – PROJECT PLANNING AND SCHEDULING

Introduction:

The Project Planning and Scheduling Phase focuses on organizing the activities required to successfully develop the *EventForce Management System*. Proper planning helps define the tasks, responsibilities, resources, timelines, and milestones involved in the project.

The project is divided into smaller and manageable activities so that each stage can be completed systematically. The schedule provides a clear timeline for completing the project from initial planning through development, testing, deployment, and documentation.

Effective planning also helps identify possible risks and provides suitable measures to minimize their impact on the project.

Work Breakdown Structure (WBS):

The *Work Breakdown Structure (WBS)* divides the EventForce Management System into smaller tasks and activities. It provides a structured view of the complete project.

EventForce Management System

1. Project Initiation

* Identify the project idea

* Identify the problem

* Define the proposed solution

* Define project objectives

2. Requirement Analysis

* Identify user requirements

* Identify functional requirements

* Identify non-functional requirements

* Identify hardware and software requirements

3. System Design

* Design system architecture

* Design modules

* Design database

* Design user interface

* Design system workflow

4. Development

* Develop frontend

* Develop backend

* Develop database

* Implement user management

* Implement event management

* Implement event registration

* Implement participant management

5. Testing

* Prepare test cases
* Perform unit testing
* Perform integration testing
* Perform system testing
* Identify and fix errors

6. Deployment
* Configure the application
* Connect the database
* Deploy the application
* Perform final verification

7. Evaluation and Documentation
* Evaluate system performance
* Verify project objectives
* Prepare final documentation
* Prepare project presentation

Task Allocation:

The project activities are assigned based on the requirements of the EventForce Management System.

Task	Responsibility
Project planning	Project Team
Requirement gathering	Project Team
System design	Project Team
Database design	Developer
Frontend development	Frontend Developer
Backend development	Backend Developer
Event management module	Developer
User management module	Developer

Registration module	Developer
Participant management	Developer
Testing	Testing Team / Developer
Error correction	Developer
Deployment	Developer
Documentation	Project Team

For an individual project, the above responsibilities can be handled by the *Project Developer* throughout the development process.

Project Milestones:

Milestones are important checkpoints used to measure the progress of the project.

Milestone 1 – Project Idea Finalization

The problem, proposed solution, objectives, and expected outcome are identified.

Milestone 2 – Requirement Completion

All major functional and non-functional requirements are identified and documented.

Milestone 3 – Design Completion

The system architecture, modules, database structure, interface, and workflow are finalized.

Milestone 4 – Development Completion

The major modules of the EventForce Management System are developed and integrated.

Milestone 5 – Testing Completion

The system is tested, and identified errors are corrected.

Milestone 6 – Deployment Completion

The completed application is configured and made ready for use.

Milestone 7 – Final Evaluation

The system is evaluated against the defined objectives and requirements.

Milestone 8 – Documentation Completion

The final project report, presentation, and supporting documentation are completed.

Resource Planning:

Resource planning identifies the resources required to develop and implement the EventForce Management System.

Human Resources

The project may require:

* Project Developer
* Frontend Developer
* Backend Developer
* Database Developer
* Tester
* Project Guide / Mentor

For an individual project, one developer can perform multiple roles.

Hardware Resources

The required hardware resources include:

* Computer or laptop
* Minimum 4 GB RAM
* Adequate storage space
* Keyboard and mouse
* Internet connection

Software Resources

The software resources may include:

* Windows / Linux operating system
* Visual Studio Code
* Web browser such as Google Chrome
* Python
* Flask or suitable backend framework
* HTML
* CSS

* JavaScript
* MySQL / SQLite
* Git/GitHub for version control, if required

Data Resources

The system requires:

* User information
* Event information
* Participant information
* Registration information
* Event schedule and venue details

Risk Management:

Risk management helps identify possible problems that may affect the development or successful completion of the project.

Project Monitoring and Control

The progress of the project will be monitored throughout the development process. Each completed activity will be compared with the planned schedule.

Project monitoring includes:

* Tracking completed and pending tasks.

* Checking development progress regularly.

* Identifying delays and technical issues.

* Testing completed modules.

* Updating the project schedule when necessary.

* Reviewing the system against the defined requirements.

* Maintaining project documentation throughout development.

Regular monitoring ensures that the project remains aligned with its objectives and planned completion date.

Expected Outcome:

The expected outcome of the Project Planning and Scheduling Phase is a clear and structured development plan for the *EventForce Management System*.

At the end of this phase:

* All major project activities will be identified.

* Tasks will be divided into manageable stages.

* Responsibilities will be clearly allocated.

* A development schedule will be established.

* Project milestones will be defined.

* Required hardware, software, human, and data resources will be identified.

* Potential project risks will be identified with appropriate mitigation strategies.

* The development team will have a clear roadmap for implementing the system.

Overall, effective project planning and scheduling will help ensure that the *EventForce Management System* is developed systematically, completed within the planned timeline, and meets the requirements identified in the earlier phases.